

Traditional Search & Rescue Collapses Across 4 Critical Voids

70% CANOPY CRASHES

15-30m GEO ERROR

2-3 HOURS DELAY

₹40L+ AIRFRAMES

EVIDENCE 01

Wayanad Canopy Crash

NDMA Official Review

DIRECT SIGNAL

GPS SIGNALS

GPS MULTI-PATH

FOREST CANOPY OBSTRUCTION

SIGNAL LOSS

SUBSYSTEM A // GNC

Canopy GPS Drift & Crashes

CORE FAILURE MECHANISM:

Dense foliage blocks satellite GNSS signals

OPERATIONAL IMPACT:

Commercial drones lose position lock, drifting into tree canopies and causing rotor loss.

70% Loss

Canopy Sorties

0 Hold

Non-GPS Stability

TARGET: PX4 & 3D ORCA

LEAD: NIKHIL

EVIDENCE 02

Chamoli Gorge RF Cut

NDRF Field Incident Report

DIGITAL CLIP

SUBSYSTEM B // COMMS

Mountain Ravine RF Blackout

CORE FAILURE MECHANISM:

Ridgelines sever direct line-of-sight RF

OPERATIONAL IMPACT:

Conventional H.264 digital video completely cuts out below 5dB SNR, causing total blindness.

<5dB SNR

Video Blackout

0 Relay

Single-Drone Link

TARGET: DEEP JSCC MESH

LEAD: NIKHIL

EVIDENCE 03

Sikkim Slope Geo Error

CAG Disaster Management Audit

35% FALSE ALARM

SUBSYSTEM C // VISION

Single-Sensor & Slope Drift

CORE FAILURE MECHANISM:

2D cameras assume flat ground; blind to smoke

OPERATIONAL IMPACT:

Smoke/darkness blind single RGB sensors; steep slopes produce 15-30m GPS coordinate errors.

15-30m

Slope Geo Drift

35%

False Alarm Rate

TARGET: TRI-MODAL & 3D DEM

LEAD: VEDANTH

EVIDENCE 04

Central Link Abort

Field Operational Bottleneck

15-25 CREW LOAD

SUBSYSTEM D // C2 GCS

Central Pilot Bottleneck

CORE FAILURE MECHANISM:

1-pilot-per-drone manual radio control

OPERATIONAL IMPACT:

Requires 15-25 personnel and 45-90 min setup; sortie aborts if the single pilot link drops.

2-3 Hrs

Search Time / mi²

₹12.5L

Cost / Deployment

TARGET: WEBGPU ATAK GCS

LEAD: SIVA

FIELD EVIDENCE

Wayanad Landslide Audit

AUDIT FINDING

70% RF/GPS Disconnect

COST PENALTY

₹40L Military Airframes

SURVIVOR RISK

412% Disaster Surge

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TEAM OFFGRID — PROJECT SUTRA